

DAS DIAL PROBLEM STATEMENT 3 (The hardest of all DAS problem statements)	
Title	DAS Adaptive Learning Activity Generator
Industry Mentors	Ms Soofrina Mubarak
Project Context	Dyslexia Association of Singapore (DAS) supports students with dyslexia and other learning differences through specialised literacy intervention programmes. Educational therapists spend significant time creating differentiated worksheets, reading passages, and learning activities tailored to individual student needs. This process is labour-intensive and difficult to scale across large student populations. DAS is exploring the use of Generative AI to support adaptive learning activity generation aligned to literacy goals, student profiles, and instructional scope-and-sequence frameworks.
Project Objectives	How can generative AI and Retrieval-Augmented Generation (RAG) techniques be utilised to: ● Generate differentiated literacy learning activities ● Align generated content with student learning profiles ● Retrieve pedagogically appropriate instructional strategies from a structured knowledge base ● Support educators in creating scalable customised learning materials ● Reduce educator preparation workload while maintaining instructional quality

Project Deliverables	Presentation with demo of prototype Prototype with the following technical requirements: • Web application integrating LLM APIs or open-source LLMs • Retrieval-Augmented Generation (RAG) pipeline • Curriculum knowledge base • Prompt engineering and content moderation pipeline • Database for generated activity storage • Evaluation framework for generated educational quality Documentation • AI architecture and prompt pipeline • RAG workflow and retrieval strategy • Safety and responsible AI considerations • Project management methodology • Evaluation and benchmarking findings
Main form of expected project deliverables	An AI-assisted adaptive learning activity generation platform
Crucial properties to test for the delivered software	Pedagogical Accuracy, Functionality, Reliability, Safety, Usability
Additional related resources	www.das.org.sg